BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: PR – XII (A-02) (21)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: Note: Four choices are given against each question; fill the circle of the correct answer only. Use a black or blue ball point / marker to fill the circle. Filling more than one circle for a question will be counted as a wrong answer.

Q1. Aldehydes or ketones are reduced to alkanes in the presence of basic medium by _____.

(A) Clemenson reduction
(B) Wolf-Kishner reduction
(C) Dow process
(D) Williamson's synthesis

Q2. Which one of the given compounds will show Silver Mirror Test.

(A) Ethanal
(B) Ethanol
(C) Ether
(D) Acetone

Q3. The more acidic compound among the given is _____.

(A) C_6H_5OH
(B) C_2H_5OH
(C) H_2CO_3
(D) CH_3COOH

Q4. The dehydrating agent which is used in preparation of acetic anhydride from acetic acid at high temperature is _____.

(A) P_2O_5
(B) H_2SO_4
(C) Al_2O_3
(D) H_3PO_4

Q5. Which one of the given is not a disaccharide?

(A) Sucrose
(B) Maltose
(C) Galactose
(D) Lactose

Q6. The hydrolysis of trilycerides by alkalies is called _____.

(A) Sponification
(B) Esterification
(C) Dehydration
(D) Trans-esterfication

Q7. Nylon (6,6) is the polymer of hexamethylene diamine and _____.

(A) Acetic acid
(B) Picric acid
(C) Adipic acid
(D) Glycol

Q8. The temperature range of troposphere is _____.

(A) −2°C to −92°C
(B) 56°C to −2°C
(C) −92°C to 1200°C
(D) 15°C to −56°C

Q9. The spectroscopic technique which is used for determination of functional group in a compound is _____.

(A) IR spectroscopy
(B) NMR
(C) UV spectroscopy
(D) Mass spectroscopy

Q10. Those elements of group IVA which have a tendency to form a dichloride arc _____.

(A) C and Si
(B) P and Si
(C) Ga and C
(D) Sn and Si

Q11. The oxidation number of cobalt in $[Co(NH_3)_4Cl_2]Cl$ is _____.

(A) 1
(B) 2
(C) 3
(D) 4

Q12. The catalyst used for the conversion of SO_2 to SO_3 in the manufacture of H_2SO_4 by contact process is _____.

(A) $TiCl_4$
(B) V_2O_5
(C) Ni
(D) Fe

Q13. Aromatic compounds are obtained upon fractional distillation of _____.

(A) Coke
(B) Coal gas
(C) Ammonical liquor
(D) Coal tar

Q14. The electrophile in the aromatic sulphonation reaction of benzene is _____.

(A) H_2SO_4
(B) SO_2
(C) SO_3
(D) H_3^+O

Q15. A special type of functional group isomerism in which the isomers are in dynamic equilibrium with each other is called _____.

(A) Metamerism
(B) Tautomerism
(C) Geometric isomerism
(D) Configurational isomerism

Q16. A two step mechanism in which a carbocation is formed in first step and in second step an unsaturated compound is formed is _____ mechanism.

(A) SN^1 (B) SN^2

(C) E^1 (D) E^2

Q17. Grignard's reagent react with acetaldehyde to form addition products which upon acid hydrolysis yield _____.

(A) Primary alcohol (B) Secondary alcohol

(C) Tertiary alcohol (D) Glycol

Q18. Picric acid is derivative of _____.

(A) Benzene (B) Benzoic acid

(C) Perchloric acid (D) Phenol

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 40 minutes. [CHEMISTRY (New) — Inter Part-II (Fresh/Reappear)]

QII. Answer any TEN parts. Each part carries FOUR marks.

1. Carbon and Silicon both belong to group IVA, but CO_2 and SiO_2 differ in their structures, comment over it.

2. Define the paramagnetic and diamagnetic substances. What feature of electronic structure is related to their behaviour?

3. Identify the functional group of following compounds. (i) $C_3H_7-C(=O)-H$ (ii) $C_2H_5-O-CH_3$ (iii) C_2H_5-COOH (iv) $(C_2H_5)_2CO$

4. How are alkenes prepared from: (i) Alcohols (ii) Alkyl halides

5. Illustrate the reason that ortho/para directing substituents activate the benzene ring while meta directing substitutents deactivate the benzene ring for further substitution reactions.

6. Compare SN^1 mechanism with E^1 mechanism.

7. What is Lucas reagent? How are different types of monohydric alcohols distinguished by using this reagent?

8. Give the conversion of aldehydes and ketones in to alkanes by Clemenson reduction and Wolf-Kishner reduction.

9. Why methanoic acid is stronger than ethanoic acid and chloro acetic acid is stronger than acetic acid?

10. Describe briefly the secondary and tertiary structures of protein.

11. Enlist four fractions that are obtained by fractional distillation of petroleum. Also give their uses.

12. Write down the harmful effects of acid rain.

13. Why are the modern methods of analysis superior over classical methods of analysis?

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 27

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [05] Define polymers. How are they classified? Explain each type by giving examples.

(b) [04] Describe the mechanism of free radical substitution reaction in halogenation of alkanes.

QIV.

(a) [05] What is the principle of UV/Vis spectroscopy? Discuss the electronic transitions that are associated with the absorption of UV/Vis radiations.

(b) [04] Beryllium differs from other members of its group. Give any four points of differences.

QV.

(a) [05] Give industrial applications of enzymes.

(b) [04] Write note on: (i) Tollen's test (ii) Williamson's synthesis

QVI.

(a) [05] Write the names of following compounds according to IUPAC system. i. CH_3–(benzene ring)–CHO (ortho-substituted benzaldehyde) ii. $[Co(NH_3)_4Cl_2]Cl$ iii. CH_3–CH_2–$CH(COOH)$–CH_2–$C(=O)$–OH iv. CH_3–CH_2–CO–$CH(CH_3)_2$

(b) [04] How will you convert: (i) Acetic acid in to ethyl acetate. (ii) Formaldehyde in to primary alcohol by using Grignard's reagent and H_2O.